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NEET · BIOLOGY · CHAPTER-WISE PYQS

# Principles of Inheritance and Variation

NEET Biology · Previous Year Questions with Answer Key

| 272 | 51 | 1988 – 2026 | English |
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## MCQ (Single Correct Answer) 272 questions

**Q1.** RE-NEET 2026

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : In an experiment, Mendel observed that the F1 progeny plants are all tall and none are dwarf.

Reason R : Stem height is a contrasting trait, with tall being dominant and dwarf being recessive.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) $\mathbf{A}$ is not correct but $\mathbf{R}$ is correct

(B) Both $\mathbf{A}$ and $\mathbf{R}$ are correct and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

(C) Both $\mathbf{A}$ and $\mathbf{R}$ are correct but $\mathbf{R}$ is not the correct explanation of $\mathbf{A}$

(D) $\mathbf{A}$ is correct but $\mathbf{R}$ is not correct

Open solution

**Q2.** RE-NEET 2026

For a person with blood group 'O', which of the following is not a possible combination of parents' blood group genotypes?

(A) Father : $I^AI^B$ and Mother : $I^Ai$

(B) Father : $I^Ai$ and Mother : $I^Ai$

(C) Father : $I^Ai$ and Mother : $I^Bi$

(D) Father : $I^Bi$ and Mother : $I^Bi$

Open solution

**Q3.** RE-NEET 2026

Given below are two statements :

Statement I : Down's syndrome is caused by the absence of one of the X-chromosomes.

Statement II : Turner's syndrome is caused by the presence of an additional copy of the chromosomes.

In the light of the above statements, choose the correct answer from the options given below:

(A) Statement I is incorrect but Statement II is correct

(B) Both Statement I and Statement II are correct

(C) Both Statement I and Statement II are incorrect

(D) Statement I is correct but Statement II is incorrect

Open solution

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**Q4.** NEET 2026

Open solution

Match List I with List II :

| List-I | | List-II | |
|---|---|---|---|
| A. | Incomplete dominance | I. | Human skin colour |
| B. | Co-dominance | II. | Inheritance of flower colour in Antirrhinum sp. |
| C. | Pleiotropy | III. | Phenylketonuria disease in humans |
| D. | Polygenic inheritance | IV. | ABO blood groups |

Choose the correct answer from the options given below :

**(A)** A-II, B-IV, C-III, D-I
**(B)** A-I, B-III, C-II, D-IV
**(C)** A-I, B-IV, C-III, D-II
**(D)** A-II, B-I, C-III, D-IV

**Q5.** NEET 2026

Open solution

Which of the following statements are true with reference to the sex-determination in honeybees?

A. An offspring formed from the union of a sperm and an egg, develops as a female (queen or worker).

B. An unfertilized egg develops as a male by parthenogenesis.

C. A male has half the number of chromosomes than that of a female.

D. Males produce sperms by meiosis.

E. Honeybees have a haplodiploid sex-determination system.

Choose the correct answer from the options given below :

**(A)** A, B, C and E only
**(B)** B, C, D and E only
**(C)** A, B, C and D only
**(D)** A, B, D and E only

**Q6.** NEET 2026

Open solution

Which one of the following disorders is caused by the substitution of Glutamic acid (Glu) by Valine (Val) at the sixth position of the beta globin chain of the haemoglobin molecule?

**(A)** Thalassemia
**(B)** Sickle-cell anaemia
**(C)** Phenylketonuria
**(D)** Haemophilia

**Q7.** NEET 2026

Open solution

What is the probability of having children with 'O' blood group, where both mother and father are heterozygous for ' $\underline{A}$ ' and ' $\underline{B}$ ' blood group, respectively?

**(A)** $25\%$ **(B)** $0\%$ **(C)** $75\%$ **(D)** $50\%$

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**Q8.** NEET 2026

In a population of a grasshopper species, the chromosome number of some members is 23 and some other members possess 24 chromosomes. The 23 and 24 chromosome-bearing members in this species are ____ .

Open solution

(A) females and males, respectively
(B) all males
(C) males and females, respectively
(D) all females

**Q9.** NEET 2025

What is the pattern of inheritance for polygenic trait?

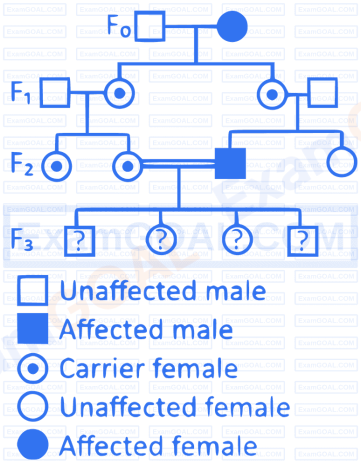
Open solution

(A) Autosomal dominant pattern
(B) X-linked recessive inheritance pattern
(C) Mendelian inheritance pattern
(D) Non-mendelian inheritance pattern

**Q10.** NEET 2025

With the help of given pedigree, find out the probability for the birth of a child having no disease and being a carrier (has the disease mutation in one allele of the gene) in $F_3$ generation.

Open solution




(A) $1/8$
(B) Zero
(C) $1/4$
(D) $1/2$

**Q11.** NEET 2025

Genes $R$ and $Y$ follow independent assortment. If RRYY produce round yellow seeds and rryy produce wrinkled green seeds, what will be the phenotypic ratio of the F2 generation?

Open solution

(A) Phenotypic ratio - $9:3:3:1$
(B) Phenotypic ratio - $9:7$
(C) Phenotypic ratio - $1:2:1$
(D) Phenotypic ratio - $3:1$

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**Q12.** NEET 2024 (Re-Examination)

Open solution

Given below are two statements:

Statement I: When many alleles of a single gene govern a character, it is called polygenic inheritance.

Statement II: In Polygenic inheritance, the effect of each allele is additive.

In the light of the above statements, choose the correct answer from the options given below.

**(A)** Statement I is true but Statement II is false

**(B)** Statement I is false but Statement II is true

**(C)** Both Statement I and Statement II are true

**(D)** Both Statement I and Statement II are false

**Q13.** NEET 2024 (Re-Examination)

Open solution

Match List-I with List-II:

| | **List - I Type of Inheritance** | | **List - II Example** |
|---|---|---|---|
| (A) | Incomplete dominance | (I) | Blood groups in human |
| (B) | Co-dominance | (II) | Flower colour in Antirrhinum |
| (C) | Pleiotropy | (III) | Skin colour in human |
| (D) | Polygenic inheritance | (IV) | Phenylketonuria |

Choose the correct answer from the options given below:

**(A)** A-III, B-IV, C-II, D-I

**(B)** A-II, B-I, C-IV, D-III

**(C)** A-II, B-III, C-I, D-IV

**(D)** A-IV, B-I, C-III, D-II

**Q14.** NEET 2024 (Re-Examination)

Open solution

Given below are two statements:

Statement I: Failure of segregation of chromatids during cell cycle resulting in the gain or loss of whole set of chromosome in an organism is known as aneuploidy.

Statement II: Failure of cytokinesis after anaphase stage of cell division results in the gain or loss of a chromosome is called polyploidy.

In the light of the above statements, choose the correct answer from the options given below:

**(A)** Statement I is true but Statement II is false

**(B)** Statement I is false but Statement II is true

**(C)** Both Statement I and Statement II are true

**(D)** Both Statement I and Statement II are false

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**Q15.** NEET 2024 (Re-Examination)

When a tall pea plant with round seeds was selfed, it produced the progeny of :

(a) Tall plants with round seeds and

(b) Tall plants with wrinkled seeds.

Identify the genotype of the parent plant.

(A) TtRr (B) TtRR (C) TTRR (D) TTRr

Open solution

**Q16.** NEET 2024 (Re-Examination)

Aneuploidy is a chromosomal disorder where chromosome number is not the exact copy of its haploid set of chromosomes, due to :

A. Substitution

B. Addition

C. Deletion

D. Translocation

E. Inversion

Choose the most appropriate answer from the options given below :

(A) C and D only (B) D and E only (C) A and B only (D) B and C only

Open solution

**Q17.** NEET 2024 (Re-Examination)

The mother has $\mathrm{A}+$ blood group the father has $\mathrm{B}+$ and the child is $\mathrm{A}+$. What can be the possibility of genotypes of all three, respectively?

A. $I^AI^A|I^Bi|I^Bi$

B. $I^AI^A\,|I^Bi|I^Ai$

C. $I^Bi\,|I^AI^A|I^AI^B$

D. $I^AI^A\,|I^BI^B|I^Ai$

E. $I^Ai\,|I^Bi|I^Ai$

Choose the correct answer from the options given below:

(A) C and D (B) D and A (C) A and B (D) B and E

Open solution

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**Q18.** NEET 2024

Open solution

Which one of the following can be explained on the basis of Mendel's Law of Dominance?

A. Out of one pair of factors one is dominant and the other is recessive.

B. Alleles do not show any expression and both the characters appear as such in $F_2$ generation.

C. Factors occur in pairs in normal diploid plants.

D. The discrete unit controlling a particular character is called factor.

E. The expression of only one of the parental characters is found in a monohybrid cross.

Choose the correct answer from the options given below:

**(A)** A, B and C only
**(B)** A, C, D and E only
**(C)** B, C and D only
**(D)** A, B, C, D and E

**Q19.** NEET 2024

Open solution

A pink flowered Snapdragon plant was crossed with a red flowered Snapdragon plant. What type of phenotype/s is/are expected in the progeny?

**(A)** Only red flowered plants
**(B)** Red flowered as well as pink flowered plants
**(C)** Only pink flowered plants
**(D)** Red, Pink as well as white flowered plants

**Q20.** NEET 2024

Open solution

In a plant, black seed color $(BB/Bb)$ is dominant over white seed color $(bb)$. In order to find out the genotype of the black seed plant, with which of the following genotype will you cross it?

**(A)** $BB$ **(B)** $bb$ **(C)** $Bb$ **(D)** $BB/Bb$

**Q21.** NEET 2024

Open solution

Match List I with List II

| | List - I | | List - II |
|---|---|---|---|
| A. | Two or more alternative forms of a gene | I. | Back cross |
| B. | Cross of $F_1$ progeny with homozygous recessive parent | II. | Ploidy |
| C. | Cross of $F_1$ progeny with any of the parents | III. | Allele |
| D. | Number of chromosome sets in plant | IV. | Test cross |

Choose the correct answer from the options given below:

**(A)** A-I, B-II, C-III, D-IV
**(B)** A-II, B-I, C-III, D-IV
**(C)** A-III, B-IV, C-I, D-II
**(D)** A-IV, B-III, C-II, D-I

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**Q22.** NEET 2024

Open solution

Match List I with List II :

| | List - I | | List - II |
|---|---|---|---|
| A. | Down's syndrome | I. | $11^{th}$ chromosome |
| B. | $\alpha$-thalassemia | II. | 'X' chromosome |
| C. | $\beta$-thalassemia | III. | $21^{st}$ chromosome |
| D. | Klinefelter's syndrome | IV. | $16^{th}$ chromosome |

Choose the correct answer from the options given below :

**(A)** A-I, B-II, C-III, D-IV

**(B)** A-II, B-III, C-IV, D-I

**(C)** A-III, B-IV, C-I, D-II

**(D)** A-IV, B-I, C-II, D-III

**Q23.** NEET 2024

Open solution

As per ABO blood grouping system, the blood group of father is $B^+$, mother is $A^+$ and child is $O^+$. Their respective genotype can be

A. $I^Bi/I^Ai/ii$

B. $I^BI^B/I^AI^A/ii$

C. $ii^B/ii^A/I^AI^B$

D. $I^Ai/I^Bi/I^Ai$

E. $ii^B/ii^A/I^AI^B$

Choose the most appropriate answer from the options given below :

**(A)** A only

**(B)** B only

**(C)** C & B only

**(D)** D & E only

**Q24.** NEET 2023 Manipur

Open solution

Match List - I with List - II.

| | List - I | | List - II |
|---|---|---|---|
| (A) | Monohybrid Cross | (I) | 1 : 1 |
| (B) | Dihybrid Cross | (II) | 1 : 2 : 1 |
| (C) | Incomplete dominance | (III) | 3 : 1 |
| (D) | Test Cross | (IV) | 9 : 3 : 3 : 1 |

Choose the correct answer from the options given below :

**(A)** (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

**(B)** (A)-(II), (B)-(IV), (C)-(III), (D)-(I)

**(C)** (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

**(D)** (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

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**Q25.** NEET 2023 Manipur

A heterozygous pea plant with violet flowers was crossed with homozygous pea plant with white flower. Violet is dominant over white. Which one of the following represents the expected combinations among 40 progenies formed?

Open solution

**(A)** 30 produced violet and 10 produced white flowers

**(B)** 20 produced violet and 20 produced white flowers

**(C)** All 40 produced violet flowers

**(D)** All 40 produced white flowers

**Q26.** NEET 2023 Manipur

In which disorder change of single base pair in the gene for beta globin chain results in change of glutamic acid to valine ?

Open solution

**(A)** Thalassemia

**(B)** Sickle cell anemia

**(C)** Haemophilia

**(D)** Phenylketonuria

**Q27.** NEET 2023 Manipur

A certain plant homozygous for yellow seeds and red flowers was crossed with a plant homozygous for green seeds and white flowers. The $F_1$ plants had yellow seeds and pink flowers. The $F_1$ plants were selfed to get $F_2$ progeny. Assuming independent assortment of the two characters, how many phenotypic categories are expected for these characters in the $F_2$ generation ?

Open solution

**(A)** 9

**(B)** 16

**(C)** 4

**(D)** 6

**Q28.** NEET 2023 Manipur

Select the correct statements about sickle cell anaemia.

(A) There is a change in gene for beta globin.

(B) In the beta globin, there is valine in the place of Lysine.

(C) It is an example of point mutation.

(D) In the normal gene U is replaced by A.

Choose the correct answer from the options given below :

Open solution

**(A)** (B), (C) and (D) only

**(B)** (B) and (D) only

**(C)** (A), (B) and (D) only

**(D)** (A) and (C) only

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**Q29.** NEET 2023 Manipur

Open solution

Select the correct statement/s with respect to mechanism of sex determination in Grasshopper.

(A) It is an example of female heterogamety.

(B) Male produces two different types of gametes either with or without X chromosome.

(C) Total number of chromosomes (autosomes and sex chromosomes) is same in both males and females.

(D) All eggs bear an additional X chromosome besides the autosomes.

Choose the **correct** answer from the options given below :

**(A)** (B) and (D) only
**(B)** (A), (C) and (D) only
**(C)** (A) only
**(D)** (A) and (C) only

**Q30.** NEET 2023

Open solution

Frequency of recombination between gene pairs on same chromosome as a measure of the distance between genes to map their position on chromosome, was used for the first time by

**(A)** Sutton and Boveri
**(B)** Alfred Sturtevant
**(C)** Henking
**(D)** Thomas Hunt Morgan

**Q31.** NEET 2023

Open solution

The phenomenon of pleiotropism refers to

**(A)** Presence of two alleles, each of the two genes controlling a single trait
**(B)** A single gene affecting multiple phenotypic expression
**(C)** More than two genes affecting a single character
**(D)** Presence of several alleles of a single gene controlling a single crossover

**Q32.** NEET 2023

Open solution

Which of the following statements are correct about Klinefelter's Syndrome?

A. This disorder was first described by Langdon Down (1866).

B. Such an individual has overall masculine development. However, the feminine development is also expressed.

C. The affected individual is short statured.

D. Physical, psychomotor and mental development is retarded.

E. Such individuals are sterile.

Choose the correct answer from the options given below :

**(A)** C and D only
**(B)** B and E only
**(C)** A and E only
**(D)** A and B only

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**Q33.** NEET 2023

Broad palm with single palm crease is visible in a person suffering from-

(A) Turner's syndrome
(B) Klinefelter's syndrome
(C) Thalassemia
(D) Down's syndrome

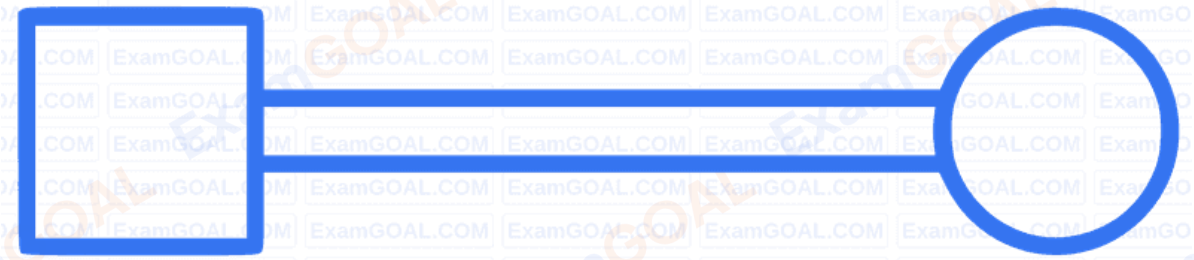
Open solution

**Q34.** NEET 2023

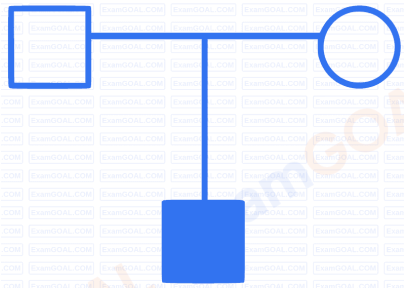
Which one of the following symbols represents mating between relatives in human pedigree analysis?

Open solution

(A)



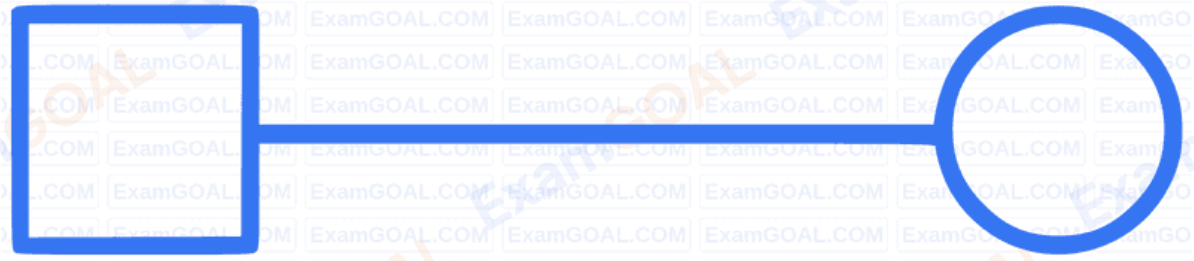
(B)



(C)



(D)



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**Q35.** NEET 2022 Phase 2

Given below are two statements:

Statement I : Sickle cell anaemia and haemophilia are autosomal dominant traits.

Statement II : Sickle cell anaemia and haemophilia are disorders of the blood.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is incorrect but Statement II is correct

(B) Both Statement I and Statement II are correct

(C) Both Statement I and Statement II are incorrect

(D) Statement I is correct but Statement II is incorrect

Open solution

**Q36.** NEET 2022 Phase 2

In meiosis, crossing over and exchange of genetic material between homologous chromosomes are catalyzed by the enzyme

(A) Polymerase (B) Phosphorylase (C) Recombinase (D) Transferase

Open solution

**Q37.** NEET 2022 Phase 2

The chromosomal theory of inheritance was proposed by

(A) Robert Brown (B) Thomas Morgan

(C) Sutton and Boveri (D) Gregor Mendel

Open solution

**Q38.** NEET 2022 Phase 2

What is the expected percentage of $F_2$ progeny with yellow and inflated pod in dihybrid cross experiment involving pea plants with green coloured, inflated pod and yellow coloured constricted pod?

(A) 9% (B) 100% (C) 56.25% (D) 18.75%

Open solution

**Q39.** NEET 2022 Phase 2

If a female individual is with small round head, furrowed tongue, partially open mouth and broad palm with characteristic palm crease. Also the physical, psychomotor and mental development is retarded. The karyotype analysis of such an individual will show :

(A) Trisomy of chromosome 21

(B) 47 chromosomes with XXY sex chromosomes

(C) 45 chromosomes with XO sex chromosomes

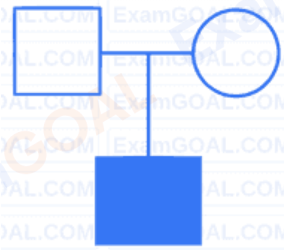
(D) 47 chromosomes with XYY sex chromosomes

Open solution

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**Q40.** NEET 2022 Phase 2

Select the incorrect match regarding the symbols used in the Pedigree analysis:

(A)  – Parent with male child affected with disease

(B)  – Sex unspecified

(C)  – Affected individual

(D)  – Consanguineous mating

Open solution

**Q41.** NEET 2022 Phase 2

A normal girl, whose mother is haemophilic marries a male with no ancestral history of haemophilia. What will be the possible phenotypes of the offspring?

(a) Haemophilic son and haemophilic daughter.

(b) Haemophilic son and carrier daughter.

(c) Normal daughter and normal son.

(d) Normal son and haemophilic daughter.

Choose the most appropriate answer from the options given below:

**(A)** (b) and (d) only **(B)** (a) and (b) only **(C)** (b) and (c) only **(D)** (a) and (d) only

Open solution

**Q42.** NEET 2022 Phase 1

XO type of sex determination can be found in :

**(A)** Drosophila **(B)** Birds **(C)** Grasshoppers **(D)** Monkeys

Open solution

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**Q43.** NEET 2022 Phase 1

Open solution

Given below are two statements:

Statement I : Mendel studied seven pairs of contrasting traits in pea plants and proposed the Laws of Inheritance.

Statement II : Seven characters examined by Mendel in his experiment on pea plants were seed shape and colour, flower colour, pod shape and colour, flower position and stem height.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are correct

(B) Both Statement I and Statement II are incorrect

(C) Statement I is correct but Statement II is incorrect

(D) Statement I is incorrect but Statement II is correct

**Q44.** NEET 2022 Phase 1

Open solution

Which of the following occurs due to the presence of autosome linked dominant trait?

(A) Sickle cell anaemia

(B) Myotonic dystrophy

(C) Haemophilia

(D) Thalessemia

**Q45.** NEET 2022 Phase 1

Open solution

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Mendel's law of independent assortment does not hold good for the genes that are located closely on the same chromosome.

Reason (R) : Closely located genes assort independently.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(B) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(C) (A) is correct but (R) is not correct

(D) (A) is not correct but (R) is correct

**Q46.** NEET 2022 Phase 1

Open solution

If a colour blind female marries a man whose mother was also colour blind, what are the chances of her progeny having colour blindness?

(A) 25%

(B) 50%

(C) 75%

(D) 100%

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**Q47.** NEET 2022 Phase 1

The recombination frequency between the genes a & c is 5%, b & c is 15%, b & d is 9%, a & b is 20%, c & d is 24% and a & d is 29%. What will be the sequence of these genes on a linear chromosome?

(A) a, d, b, c (B) d, b, a, c (C) a, b, c, d (D) a, c, b, d

Open solution

**Q48.** NEET 2021

The production of gametes by the parents, formation of zygotes, the $F_1$ and $F_2$ plants, can be understood from a diagram called :

(A) Net square (B) Bullet square (C) Punch square (D) Punnett square

Open solution

**Q49.** NEET 2021

In a cross between a male and female, both heterozygous for sickle cell anaemia gene, what percentage of the progeny will be diseased?

(A) 100% (B) 50% (C) 75% (D) 25%

Open solution

**Q50.** NEET 2020 Phase 2

Chromosomal theory of inheritance was proposed by :

(A) Watson and Crick (B) Sutton and Boveri
(C) Bateson and Punnet (D) T. H. Morgan

Open solution

**Q51.** NEET 2020 Phase 2

The number of contrasting characters studied by Mendel for his experiments was :

(A) 7 (B) 14 (C) 4 (D) 2

Open solution

**Q52.** NEET 2020 Phase 2

The best example for pleiotropy is :

(A) ABO Blood group (B) Skin colour (C) Phenylketoneuria (D) Colour Blindness

Open solution

**Q53.** NEET 2020 Phase 1

Experimental verification of the chromosomal theory of inheritance was done by

(A) Sutton (B) Mendel (C) Morgan (D) Boveri

Open solution

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**Q54.** NEET 2020 Phase 1

Select the correct match

(A) Thalassemia – X linked

(B) Sickle cell anaemia – Autosomal recessive trait, chromosome-11

(C) Phenylketonuria – Autosomal dominant trait

(D) Haemophilia – Y linked

Open solution

**Q55.** NEET 2020 Phase 1

Identify the wrong statement with reference to the gene 'I' that controls ABO blood groups.

(A) Allele 'i' does not produce any sugar.

(B) The gene (I) has three alleles.

(C) When $I^A$ and $I^B$ are present together, they express same type of sugar.

(D) A person will have only two of the three alleles.

Open solution

**Q56.** NEET 2020 Phase 1

How many true breeding pea plant varieties did Mendel select as pairs, which were similar except in one character with contrasting traits?

(A) 14 (B) 4 (C) 2 (D) 8

Open solution

**Q57.** NEET 2019 (Odisha)

Match the items of Column-I with Column-II:

| Column-I | Column-II |
|---|---|
| (a) XX-XO method of Sex Determination | (i) Turner's Syndrome |
| (b) XX-XY method of Sex Determination | (ii) Female Heterogametic |
| (c) Karyotype-45 | (iii) Grasshopper |
| (d) ZW-ZZ method of Sex Determination | (iv) Female homogametic |

Select the correct option from the following:

(A) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)

(B) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

(C) (a)-(i), (b)-(iv), (c)-(ii), (d)-(iii)

(D) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

Open solution

**Q58.** NEET 2019 (Odisha)

In which genetic condition, each cell in the affected person, has three sex chromosomes XXY?

(A) Turner's Syndrome

(B) Thalassemia

(C) Kleinfelter's Syndrome

(D) Phenylketonuria

Open solution

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**Q59.** NEET 2019 (Odisha)

The production of gametes by the parents, the formation of zygotes, the $F_1$ and $F_2$ plants, can be understood using

(A) Wenn diagram
(B) Pie diagram
(C) A pyramid diagram
(D) Punnet square

Open solution

**Q60.** NEET 2019 (Odisha)

In a marriage between male with blood group A and female with blood group B, the progeny had either blood group AB or B. What could be the possible genotype of parents?

(A) $I^A i$ (Male) ; $I^B i$ (Female)
(B) $I^A i$ (Male) ; $I^B I^B$ (Female)
(C) $I^A I^A$ (Male) ; $I^B I^B$ (Female)
(D) $I^A I^A$ (Male) ; $I^B i$ (Female)

Open solution

**Q61.** NEET 2019

The frequency of recombination between gene pairs on the same chromosome as a measure of the distance between genes was explained by :

(A) Alfred Sturtevant
(B) Gregor J. Mendel
(C) Sutton Boveri
(D) T.H. Morgan

Open solution

**Q62.** NEET 2019

In *Antirrhinum* (Snapdragon), a red flower was crossed with a white flower and in $F_1$ generation, pink flowers were obtained. When pink flowers were selfed, the $F_2$ generation showed white, red and pink flowers. Choose the **incorrect** statement from the following :

(A) Ratio of $F_2$ is $\frac{1}{4}$ (Red) : $\frac{2}{4}$ (Pink) : $\frac{1}{4}$ (White)
(B) Law of Segregation does not apply in this experiment
(C) This experiment does not follow the Principle of Dominance
(D) Pink colour in $F_1$ is due to incomplete dominance

Open solution

**Q63.** NEET 2019

What map unit (Centimorgan) is adopted in the construction of genetic maps?

(A) A unit of distance between genes on chromosomes, representing 1% cross over.
(B) A unit of distance between genes on chromosomes, representing 50% cross over.
(C) A unit of distance between two expressed genes, representing 10% cross over.
(D) A unit of distance between two expressed genes, representing 100% cross over.

Open solution

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**Q64.** NEET 2019

What is the genetic disorder in which an individual has an overall masculine development, gynaecomastia, and is sterile?

(A) Edward syndrome
(B) Down's syndrome
(C) Turner's syndrome
(D) Klinefelter's syndrome

Open solution

**Q65.** NEET 2019

Select the **incorrect** statement.

(A) In domesticated fowls, sex of progeny depends on the type of sperm rather than egg.
(B) Human males have one of their sex-chromosome much shorter than the other.
(C) Male fruit fly is heterogametic
(D) In male grasshoppers, 50% of sperms have no sex-chromosome.

Open solution

**Q66.** NEET 2018

Which of the following pairs is wrongly matched?

(A) Starch synthesis in pea : Multiple alleles
(B) ABO blood grouping : Co-dominance
(C) XO type sex determination: Grasshopper
(D) T.H. Morgan : Linkage

Open solution

**Q67.** NEET 2018

Which of the following characteristics represent 'Inheritance of blood groups' in humans?
a. Dominance
b. Co-dominance
c. Multiple allele
d. Incomplete dominance
e. Polygenic inheritance

(A) b, c and e
(B) a, b and c
(C) b, d and e
(D) a, c and e

Open solution

**Q68.** NEET 2018

A woman has an X-linked condition on one of her X chromosomes. This chromosome can be inherited by

(A) only daughters
(B) only sons
(C) only grandchildren
(D) both sons and daughters

Open solution

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**Q69.** NEET 2017

Thalassemia and sickle cell anemia are caused due to a problem in globin molecule synthesis. Select the correct statement.

(A) Both are due to a qualitative defect in globin chain synthesis

(B) Thalassemia is due to less synthesis of globin molecules

(C) Sickle cell anemia is due to a quantitative problem of globin molecules

(D) Both are due to a quantitative defect in globin chain synthesis

Open solution

**Q70.** NEET 2017

The genotypes of a Husband and Wife are $I^AI^B$ and $I^Ai$.
Among the blood types of their children, how many different genotypes and phenotypes are possible?

(A) 4 genotypes ; 3 phenotypes

(B) 4 genotypes ; 4 phenotypes

(C) 3 genotypes ; 4 phenotypes

(D) 3 genotypes ; 3 phenotypes

Open solution

**Q71.** NEET 2017

A disease caused by an autosomal primary non-disjunction is :

(A) Klinefelter's Syndrome

(B) Down's Syndrome

(C) Turner's Syndrome

(D) Sickle Cell Anemia

Open solution

**Q72.** NEET 2017

Which one from those given below is the period for Mendel's hybridisation experiments ?

(A) 1870 – 1877

(B) 1857 – 1869

(C) 1840 – 1850

(D) 1856 – 1863

Open solution

**Q73.** NEET 2017

Among the following characters, which one was not considered by Mendel in his experiments on pea?

(A) Stem-Tall or dwarf

(B) Pod-Inflated or constricted

(C) Trichomes-Glandular or non-glandular

(D) Seed-Green or yellow

Open solution

**Q74.** NEET 2016 Phase 2

The mechanism that causes a gene to move from one linkage group to another is called

(A) crossing-over

(B) translocation

(C) duplication

(D) inversion

Open solution

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**Q75.** NEET 2016 Phase 2

If a colour-blind man marries a woman who is homozygous for normal colour vision, the probability of their son being colour-blind is

(A) 0.75 (B) 0.5 (C) 0 (D) 1

Open solution

**Q76.** NEET 2016 Phase 1

A cell at telophase stage is observed by a student in a plant brought from the field. He tells his teacher that this cell is not like other cells at telophase stage. There is no formation of cell plate and thus the cell is contaning more number of chromosomes as compared to other dividing cells. This would results in -

(A) Polyteny
(B) Aneuploidy
(C) Polyploidy
(D) Somaclonal variation

Open solution

**Q77.** NEET 2016 Phase 1

Pick out the **correct** statements :
(a) Haemophilia is a sex-linked recessive disease.
(b) Down's syndrome is due to aneuploidy.
(c) Phenylketonuria is an autosomal recessive gene disorder.
(d) Sickle cell anaemia is an X – linked recessive gene disorder.

(A) (a), (b) and (c) are correct
(B) (a), (c) and (d) are correct.
(C) (a) and (d) are correct
(D) (b) and (d) are correct.

Open solution

**Q78.** NEET 2016 Phase 1

A tall true breeding garden pea plant is crossed with a dwarf true breeding garden pea plant. When the $F_1$ plants were selfed the resulting genotypes were in the ratio of -

(A) 3 : 1 :: Tall : Dwarf
(B) 1 : 2 : 1 :: Tall heterozygous : Tall homozygous : Dwarf
(C) 1 : 2 : 1 :: Tall homozygous : Tall heterozygous : Dwarf
(D) 3 : 1 :: Dwarf : Tall

Open solution

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**Q79.** NEET 2016 Phase 1

Open solution

Match the terms in Column-I with their description in Column-II and choose the correct option :

| Column-I | Column-II |
|---|---|
| (a) Dominance | (i) Many genes govern a single character |
| (b) Codominance | (ii) In a heterozygous organism only one allele expresses itself |
| (c) Pleiotropy | (iii) In a heterozygous organism both alleles express themselves fully |
| (d) Polygenic inheritance | (iv) A single gene influences many characters |

(A) (a) - (ii), (b) - (iii), (c) - (iv), (d) - (i)

(B) (a) - (iv), (b) - (i), (c) - (ii), (d) - (iii)

(C) (a) - (ii), (b) - (ii), (c) - (iv), (d) - (iii)

(D) (a) - (iv), (b) - (iii), (c) - (i), (d) - (ii)

**Q80.** NEET 2016 Phase 1

Open solution

In a testcross involving $F_1$ dihybrid flies, more parental-type offspring were produced than the recombinant-typeoffspring. This indicates

(A) Both of the characters are controlled by more than one gene

(B) The two genes are linked and present on the same chromosome

(C) The two genes are located on two different chromosomes

(D) Chromosomes failed to separate during meiosis

**Q81.** NEET 2016 Phase 1

Open solution

Which of the following most appropriately describes haemophilia ?

(A) Chromosomal disorder

(B) X-linked recessive gene disorder

(C) Dominant gene disorder

(D) Recessive gene disorder

**Q82.** AIPMT 2015

Open solution

A gene showing codominance has :

(A) alleles tightly linked on the same chromosome

(B) one allele dominant on the other

(C) both alleles independently expressed in the heterozygote

(D) alleles that are recessive to each other

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**Q83.** AIPMT 2015

In his classic experiments on pea plants, Mendel did not use :

(A) Seed colour (B) Flower position (C) Seed shape (D) Pod length

Open solution

**Q84.** AIPMT 2015

A colour blind man marries a woman with normal sight who has no history of colour blindness in her family. What is the probability of their grandson being colour blind?

(A) 1 (B) Nil (C) 0.25 (D) 0.5

Open solution

**Q85.** AIPMT 2015

A pleiotropic gene -

(A) is a gene evolved during Pliocene

(B) is expressed only in primitive plants

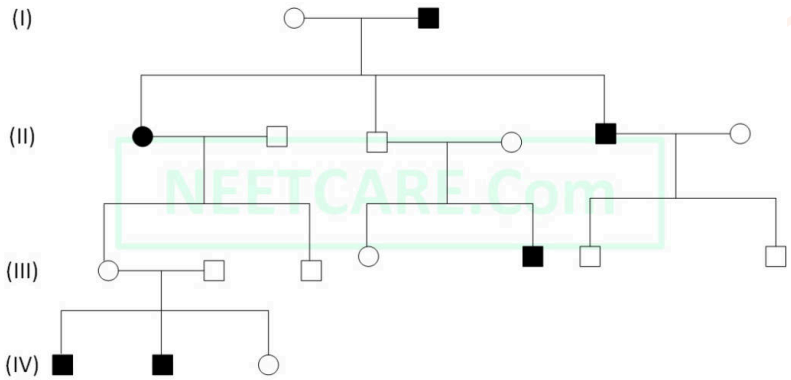
(C) Controls multiple traits in an individual

(D) controls a trait only in combination with another gene

Open solution

**Q86.** AIPMT 2015

In the following human pedigree, the filled symbols represent the affected individuals. Identify the type of given pedigree.




(A) X-linked dominant (B) Autosomal dominant

(C) Autosomal recessive (D) X-linked recessive

Open solution

**Q87.** AIPMT 2015

The term "linkage" was coined by -

(A) G. Mendel (B) T.H. Morgan (C) W. Sutton (D) T. Boveri

Open solution

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**Q88.** AIPMT 2015 Cancelled Paper

The movement of a gene from one linkage group to another is called

(A) inversion (B) translocation (C) duplication (D) crossing over

Open solution

**Q89.** AIPMT 2015 Cancelled Paper

Alleles are

(A) true breeding homozygotes
(B) different phenotype
(C) heterozygotes
(D) different molecular forms of a gene

Open solution

**Q90.** AIPMT 2015 Cancelled Paper

Multiple alleles are present

(A) on different chromosomes
(B) at the same locus of the chromosome
(C) at different loci on the same chromosome.
(D) on non-sister chromatids

Open solution

**Q91.** AIPMT 2015 Cancelled Paper

A man with blood group 'A' marries a woman with blood group 'B'. What are all the possible blood groups of their offsprings?

(A) A and B only (B) A, B, AB and O (C) A, B and AB only (D) O only

Open solution

**Q92.** AIPMT 2015 Cancelled Paper

How many pairs of contrasting characters in pea plants were studied by Mendel in his experiments?

(A) Eight (B) Five (C) Seven (D) Six

Open solution

**Q93.** AIPMT 2015 Cancelled Paper

An abnormal human baby with 'XXX' sex chromosomes was born due to

(A) formation of abnormal sperms in the father
(B) fusion of two ova and one sperm
(C) formation of abnormal ova in the mother
(D) fusion of two sperms and one ovum

Open solution

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**Q94.** AIPMT 2014

Fruit colour in squash is an example of :

(A) Recessive epistasis
(B) Dominant epistasis
(C) Complementary genes
(D) Inhibitory genes

Open solution

**Q95.** AIPMT 2014

A man whose father was colour blind marries a woman who had a colour blind mother and normal father. What percentage of male children of this couple will be colour blind?

(A) 50%
(B) 25%
(C) 0%
(D) 75%

Open solution

**Q96.** AIPMT 2014

A human female with Turner's syndrome :

(A) is able to produce children with normal husband.
(B) has 45 chromosomes with XO.
(C) exhibits male characters.
(D) has one additional X chromosome.

Open solution

**Q97.** NEET 2013 (Karnataka)

In our society women are blamed for producing female children. Choose the correct answer for the sex-determination in humans.

(A) Due to some defect like aspermia in man
(B) Due to the genetic make up of the particular sperm which fertilizes the egg
(C) Due to the genetic make up of the egg
(D) Due to some defect in the women

Open solution

**Q98.** NEET 2013 (Karnataka)

Which one is the incorrect statement with regards to the importance of pedigree analysis?

(A) It confirms that DNA is the carrier of genetic information
(B) It helps to trace the inheritance of a specific trait
(C) It confirms that the trait is linked to one of the autosome
(D) It helps to understand whether the trait in question is dominant or recessive

Open solution

**Q99.** NEET 2013 (Karnataka)

Down's syndrome in humans is due to :

(A) Three 'X' chromosomes
(B) Monosomy
(C) Three copies of chromosome 21
(D) Two 'Y' chromosomes

Open solution

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**Q100.** NEET 2013

Select the incorrect statement with regard to Haemophilia :

(A) It is a dominant disease

(B) A single protein involved in the clotting of blood is affected

(C) It is a recessive disease

(D) It is a sex-linked disease

Open solution

**Q101.** NEET 2013

Which of the following cannot be detected in a developing foetus by amniocentesis ?

(A) Sex of the foetus

(B) jaundice

(C) Klinefelter syndrome

(D) Down syndrome

Open solution

**Q102.** NEET 2013

If both parents are carriers for thalessemia, which is an autosomal recessive disorder, what are the chances of pregnancy resulting in an affected child ?

(A) 100%

(B) 25%

(C) no chance

(D) 50%

Open solution

**Q103.** NEET 2013

If two persons with 'AB' blood group marry and have sufficiently large number of children, these children could be classified as 'A' blood group : 'AB' blood group : 'B' blood group in 1 : 2 : 1 ratio. Modern technique of protein electrophoresis reveals presence of both 'A' and 'B' type proteins in 'AB' blood group individuals. This is an example of :

(A) Codominance

(B) Partial dominance

(C) Incomplete dominance

(D) Complete dominance

Open solution

**Q104.** NEET 2013

Which idea is depicted by a cross in which the $F_1$ generation resembles both the parents?

(A) law of dominance

(B) co-dominance

(C) incomplete dominance

(D) inheritance of one gene

Open solution

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**Q105.** NEET 2013

Which of the following statements is not true of two genes that show 50% recombination frequency ?

Open solution

(A) The genes are tightly linked

(B) The genes show independent assortment

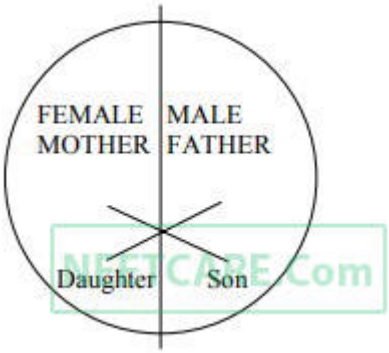
(C) If the genes are present on the same chromosome, they undergo more than one crossovers in every meiosis.

(D) The genes may be on different chromosomes

**Q106.** AIPMT 2012 Mains

Represented below is the inheritance pattern of a certain type of traits in humans. Which one of the following conditions could be an example of this pattern ?

Open solution




(A) Haemophilia

(B) Sickle cell anaemia

(C) Thalassemia

(D) Phenylketonuria

**Q107.** AIPMT 2012 Mains

A test cross is carried out to :

Open solution

(A) Determine whether two species or varieties will breed successfully

(B) Determine the genotype of a plant at $F_2$

(C) Predict whether two traits are linked

(D) Assess the number of alleles of a gene

**Q108.** AIPMT 2012 Mains

Which one of the following is a wrong statement regarding mutations ?

Open solution

(A) UV and Gamma rays are mutagens

(B) Deletion and insertion of base pairs cause frame-shift mutations

(C) Cancer cells commonly show chromosomal aberrations

(D) Change in a single base pair of DNA does not cause mutation

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**Q109.** AIPMT 2012 Prelims

A normal - visioned man whose father was colour blind, marries a woman whose father was also colour - blind. They have their first child as a daughter. What are the chances that this child would be colour blind

Open solution

(A) 50% (B) 25% (C) 100% (D) Zero percent

**Q110.** AIPMT 2012 Prelims

$F_2$ generation in a Mendelian cross showed that both genotypic and phenotypic ratios are same as 1 : 2 : 1. It represents a case of

Open solution

(A) Monohybrid cross with incomplete dominance

(B) Co - dominance

(C) Monohybrid cross with complete dominance

(D) Dihybrid cross

**Q111.** AIPMT 2011 Mains

Which one of the following conditions of the zygotic cell would lead to the birth of a normal human female child ?

Open solution

(A) only one Y chromosome (B) only one X chromosome

(C) one X and one Y chromosome (D) two X chromosomes

**Q112.** AIPMT 2011 Mains

Test cross in plants or in Drosophila involves crossing :

Open solution

(A) the $F_1$ hybrid with a double recessive genotype

(B) between two genotypes with dominant trait

(C) between two $F_1$ hybrids

(D) between two genotypes with recessive trait

**Q113.** AIPMT 2011 Prelims

When two unrelated individuals or lines are crossed, the performance of F1 hybrid is often superior to both parents. This phenomenon is called :

Open solution

(A) Splicing (B) Transformation (C) Heterosis (D) Metamorphosis

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**Q114.** AIPMT 2011 Prelims

Which one of the following conditions correctly describes the manner of determining the sex ?

(A) XO condition in humans as found in Turner Syndrome, determines female sex

(B) XO type of sex chromosomes determine male sex in grasshopper

(C) Homozygous sex chromosomes (ZZ) determine female sex in Birds

(D) Homozygous sex chromosomes (XX) produce male in Drosophila

Open solution

**Q115.** AIPMT 2011 Prelims

Mutations can be induced with :

(A) Gamma radiation

(B) Infra Red radiations

(C) IAA

(D) Ethylene

Open solution

**Q116.** AIPMT 2010 Mains

ABO blood grouping is controlled by gene I which has three alleles and show co-dominance. There are six genotypes. How many phenotypes in all are possible -

(A) five

(B) four

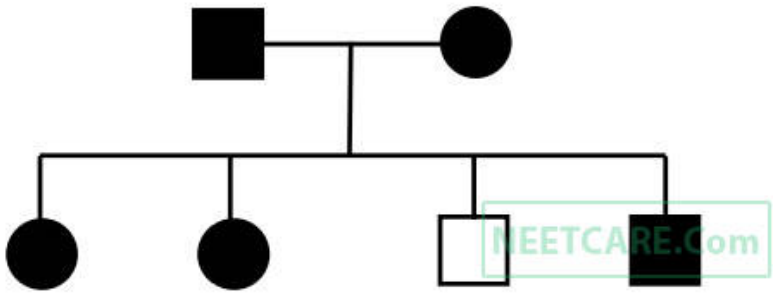
(C) three

(D) six

Open solution

**Q117.** AIPMT 2010 Mains

Study the pedigree chart of a certain family given below and select the correct conclusion which can be drawn for the character –




(A) The trait under study could not be colourblindness

(B) The female parent is heterozygous

(C) The parents could not have had a normal daughter for this character

(D) The male parent is homozygous dominant

Open solution

**Q118.** AIPMT 2010 Mains

A cross in which an organism showing a dominant phenotype in crossed with the recessive parent in order to know its genotype is called -

(A) Dihybrid cross

(B) Back cross

(C) Monohybrid cross

(D) Test cross

Open solution

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**Q119.** AIPMT 2010 Mains

In antirrhinum two plants with pink flowers were hybridized. The $F_1$ plants produced red, pink and white flowers in the proportion of 1 red, 2 pink and 1 white. What could be the genotype of the two plants used for hybridization ? Red flower colour is determined by RR, and White by rr genes -

(A) rrrr (B) rr (C) Rr (D) RR

Open solution

**Q120.** AIPMT 2010 Mains

The fruit fly *Drosophila melanogaster* was found to be very suitable for experimental verification of chromosomal theory of inheritance by Morgan and his colleagues because

(A) It reproduces parthenogenetically

(B) A single mating produces two young flies

(C) Smaller female is easily recognisable from larger male

(D) It completes life cycle in about two weeks

Open solution

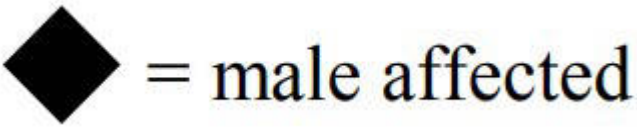
**Q121.** AIPMT 2010 Prelims

Which one of the following symbols and its representation, used in human pedigree analysis is correct –

(A) 


(B) 


(C) 


(D) 


Open solution

**Q122.** AIPMT 2010 Prelims

ABO blood groups in humans are controlled by the gene I. It has three alleles – $I^A$, $I^B$ and i. Since there are three different alleles, six different genotypes are possible. How many phenotypes can occur –

(A) One (B) Two (C) Three (D) Four

Open solution

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**Q123.** AIPMT 2010 Prelims

Select the correct statement from the ones given below with respect to dihybrid cross –

(A) Genes far apart on the same chromosome show very few recombinations

(B) Tightly linked genes on the same chromosome show very few recombination

(C) Tightly linked genes on the same chromosome show higher recombinations

(D) Genes loosely linked on the same chromosome show similar recombinations as the tightly linked ones

Open solution

**Q124.** AIPMT 2010 Prelims

The genotype of a plant showing the dominant phenotype can be determined by –

(A) Pedigree analysis

(B) Back cross

(C) Test cross

(D) Dihybrid cross

Open solution

**Q125.** AIPMT 2010 Prelims

Which one of the following cannot be explained on the basis of Mendel's Law of Dominance?

(A) Out of one pair of factors one is dominant, and the other recessive

(B) Alleles do not show any blending and both the characters recover as such in $F_2$ generation

(C) The discrete unit controlling a particular character is called a factor

(D) Factors occur in pairs

Open solution

**Q126.** AIPMT 2009

Select the incorrect statement from the following :

(A) Baldness is a sex limited trait

(B) Galactosemia is an inborn error of metabolism

(C) Linkage is an exception to the principle of independent assortment in heredity.

(D) Small population size results in random genetic drift in a population

Open solution

**Q127.** AIPMT 2009

Sickle-cell anemia is :

(A) Caused by substitution of valine by glutamic acid in the beta globin chain of haemoglobin

(B) An autosomal linked dominant trait

(C) Caused by a change in a single base pair of DNA

(D) Characterized by elongated sickle like RBCs with a nucleus

Open solution

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**Q128.** AIPMT 2009

The most popularly known blood grouping is the ABO grouping. It is named ABO and not ABC, because "O" in it refers to having :

Open solution

(A) Other antigens besides A and B on RBCs

(B) No antigens A and B on RBCs

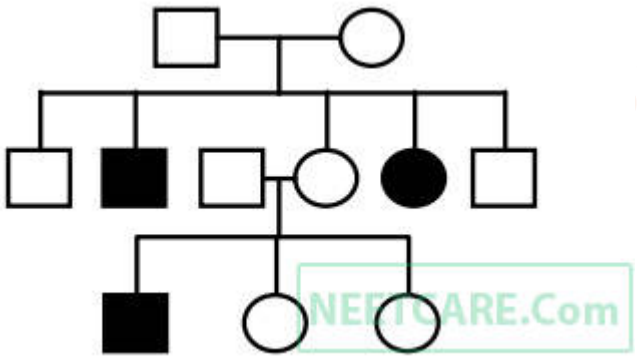
(C) Overdominance of this type on the genes for A and B types

(D) One antibody only-either anti-A or anti-B on the RBCs

**Q129.** AIPMT 2009

Study the pedigree chart given below :

Open solution




What does it show ?

(A) Inheritance of a recessive sex-linked disease like haemophilia

(B) Inheritance of a sex-linked inborn error of metabolism like phenylketonuria

(C) The pedigree chart is wrong as this is not possible

(D) Inheritance of a condition like phenylketonuria as an autosomal recessive trait

**Q130.** AIPMT 2009

Point mutation involves:

Open solution

(A) Change in single base pair

(B) Duplication

(C) Deletion

(D) Insertion

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**Q131.** AIPMT 2008

Which one of the following conditions in humans is correctly matched with its chromosomal abnormality/linkage?

(A) Klinfelter's syndrome — 44 autosomes + XXY

(B) Colour-blindness — Y-linked

(C) Down syndrome — 44 autosomes + XO

(D) Erythroblastosis foetalis — X-linked

Open solution

**Q132.** AIPMT 2008

Haploids are more suitable for mutation studies than the diploids. This is because:

(A) all mutations, whether dominant or recessive are expressed in haploids

(B) mutagens penetrate in haploids more effectively than in diploids

(C) haploids are reproductively more stable than diploids

(D) haploids are more abundant in nature than diploids

Open solution

**Q133.** AIPMT 2007

A human male produces sperms with the genotypes AB, Ab, aB, and ab pertaining to two diallelic characters in equal proportions. What is the corresponding genotype of this person ?

(A) AaBb (B) AABB (C) AaBB (D) AABb

Open solution

**Q134.** AIPMT 2007

In pea plants, yellow seeds are dominant to green. If a heterozygous yellow seeded plant is crossed with a green seeded plant, what ratio of yellow and green seeded plants would you expect in $F_1$ generation ?

(A) 9 : 1 (B) 3 : 1 (C) 50 : 50 (D) 1 : 3

Open solution

**Q135.** AIPMT 2007

Inheritance of skin colour in humans is an example of:-

(A) Chromosomal aberration (B) Polygenic inheritance

(C) Codominance (D) Point mutation

Open solution

**Q136.** AIPMT 2007

In the hexaploid wheat, the haploid(n) and basic(x) numbers of chromosomes are :

(A) n = 21 and x = 14 (B) n = 21 and x = 21

(C) n = 7 and x = 21 (D) n = 21 and x = 7

Open solution

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**Q137.** AIPMT 2007

A common test to find the genotype of a hybrid is by:-

(A) Crossing of one $F_1$ progeny with male parent

(B) Crossing of one $F_2$ progeny with female parent

(C) Crossing of one $F_2$ progeny with male parent

(D) Studying the sexual behaviour of $F_1$ progenies

Open solution

**Q138.** AIPMT 2007

Two genes R and Y are located very close on the chromosomal linkage map of maize plant. When RRYY and rryy genotypes are hybridized, the $F_2$ segregation will show

(A) segregation in the expected 9 : 3 : 3 : 1 ratio

(B) segregation in 3 : 1 ratio

(C) higher number of the parental types

(D) higher number of the recombinant types

Open solution

**Q139.** AIPMT 2006

Both sickle cell anemia and Huntington's chorea are-

(A) Congenital disorders

(B) Bacteria-related diseases

(C) Virus-related diseases

(D) Pollutant-induced disorders

Open solution

**Q140.** AIPMT 2006

Test cross involves-

(A) Crossing between two $F_1$ hybrids

(B) Crossing the $F_1$ hybrid with a double recessive genotype

(C) Crossing between two genotypes with recessive trait

(D) Crossing between two genotypes with dominant trait

Open solution

**Q141.** AIPMT 2006

If a colour blind woman marries a normal visioned man, their sons will be

(A) All normal visioned

(B) Three-fourths colourblind and one-fourth normal

(C) One-half colourblind and one-half normal

(D) All colourblind

Open solution

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**Q142.** AIPMT 2006

Cri-du-chat syndrome in humans is caused by the-

(A) Loss of half of the long arm of chromosome 5

(B) Loss of half of the short arm of chromosome 5

(C) Trisomy of $21^{st}$ chromosome

(D) Fertilization of an XX egg by a normal Y-bearing sperm

Open solution

**Q143.** AIPMT 2006

Sickle cell anaemia has not been eliminated from the African population because-

(A) It is not a fatal disease

(B) It is controlled by dominant genes

(C) It is controlled by recessive genes

(D) It provides immunity against malaria

Open solution

**Q144.** AIPMT 2006

In Mendel's experiments with garden pea, round seed shape (RR) was dominant over wrinkled seeds (rr), yellow cotyledon (YY) was dominant over green cotyledon (yy). What are the expected phenotypes in the $F_2$ generation of the cross RRYY × rryy ?

(A) Round seeds with yellow cotyledons, and wrinkled seeds with yellow cotyledons

(B) Only wrinkled seeds with green cotyledons

(C) Only wrinkled seeds with yellow cotyledons

(D) Only round seeds with green cotyledons

Open solution

**Q145.** AIPMT 2006

How many different kinds of gametes will be produced by a plant having the genotype AABbCC ?

(A) Two

(B) Nine

(C) Four

(D) Three

Open solution

**Q146.** AIPMT 2006

Phenotype of an organism is the result of-

(A) Environmental changes and sexual dimorphism

(B) Cytoplasmic effects and nutrition

(C) Genotype and environment interactions

(D) Mutations and linkages

Open solution

**Q147.** AIPMT 2006

Which one of the following is an example of polygenic inheritance ?

(A) Skin colour in humans

(B) Pod shape in garden pea

(C) Production of male honey bee

(D) Flower colour in Mirabilis jalapa

Open solution

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**Q148.** AIPMT 2006

Which one of the following is the most suitable medium for culture of *Drosophila melanogaster*?

(A) Moist bread (B) Agar agar (C) Ripe banana (D) Cow dung

Open solution

**Q149.** AIPMT 2006

In which mode of inheritance do you expect more maternal influence among the offspring?

(A) Autosomal (B) Cytoplasmic (C) Y-linked (D) X-linked

Open solution

**Q150.** AIPMT 2005

In order to find out the different types of gametes produced by a pea plants having the genotype AaBb, it should be crossed to a plant with the genotype -

(A) aaBB (B) aabb (C) AaBb (D) AABB

Open solution

**Q151.** AIPMT 2005

G-6-P dehydrogenase deficiency is associated with haemolysis of -

(A) RBCs (B) Lymphocytes (C) Platelets (D) Leucocytes

Open solution

**Q152.** AIPMT 2005

A man and a woman, who do not show any apparent signs of a certain inherited disease, have seven children (2 daughters and 5 sons). Three of the sons suffer from the given disease but none of the daughters are affected. Which of the following mode of inheritance do you suggest for this disease

(A) Autosomal dominant (B) Sex-linked recessive
(C) Sex-linked dominant (D) Sex-limited recessive

Open solution

**Q153.** AIPMT 2005

A women with 47 chromosomes due to three copies of chromosome 21 is characterized by -

(A) Superfemaleness (B) Turner's syndrome
(C) Triploidy (D) Down's syndrome

Open solution

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**Q154.** AIPMT 2005

Haemophilia is more commonly seen in human males than in human females because -

(A) This disease is due to an X-linked dominant mutation

(B) This disease is due to a Y-linked recessive mutation

(C) This disease is due to an X-linked recessive mutation

(D) A greater proportion of girls die in infancy

Open solution

**Q155.** AIPMT 2005

Which of the following is not a hereditary disease ?

(A) Cretinism (B) Cystic fibrosis (C) Haemophilia (D) Thalasasemia

Open solution

**Q156.** AIPMT 2005

A woman with normal vision, but whose father was colour blind, marries a colour blind man. Suppose that the fourth child of this couple was a boy. This boy -

(A) May be colour blind or may be normal vision

(B) Must have normal colour vision

(C) Must be colour blind

(D) Will be partially colour blind since he is heterozygous for the colour blind mutant allele.

Open solution

**Q157.** AIPMT 2005

Genes for cytoplasmic male sterility in plants are generally located in

(A) mitochondrial genome (B) cytosol

(C) chloroplast genome (D) nuclear genome

Open solution

**Q158.** AIPMT 2005

The salivary gland chromosomes in the dipteran larvae, are useful in gene mapping because

(A) these are much longer in size (B) these are easy to stain

(C) these are fused (D) they have endoreduplicated chromosomes

Open solution

**Q159.** AIPMT 2004

In a plant, red fruit (R) is dominant over yellow fruit (r) and tallness (T) is dominant over shortness (t). If a plant with RRTt genotype is crossed with a plant that is rrtt

(A) 75% will be tall with red fruit

(B) 25% will be tall with red fruit

(C) 50% will be tall with red fruit

(D) All the offspring will be tall with red fruit

Open solution

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**Q160.** AIPMT 2004

A male human is heterozygous for autosomal genes A and B and is also hemizygous for hemophilic gene h. What proportion of his sperms will be abh :-

(A) 1/16 (B) 1/32 (C) 1/8 (D) 1/4

Open solution

**Q161.** AIPMT 2004

Lack of independent assortment of two genes A and B in fruit fly Drosophila is due to :-

(A) Linkage (B) Recombination (C) Repulsion (D) Crossing over

Open solution

**Q162.** AIPMT 2004

One of the parents of a cross has a mutation in its mitochondria. In that cross, that parent is taken as a male. During segregation of $F_2$ progenies that mutation is found in -

(A) 50% of the progenies (B) 1/3 of the progenies
(C) None of the progenies (D) All the progenies

Open solution

**Q163.** AIPMT 2004

A normal woman, whose father was colour-blind is married to a normal man. The sons would be :-

(A) All colour-blind (B) 75 % colour-blind
(C) All normal (D) 50% colour-blind

Open solution

**Q164.** AIPMT 2004

The recessive genes located on X-chromosomes in humans are always-

(A) Expressed in females (B) Sub-lethal
(C) Expressed in males (D) Lethal

Open solution

**Q165.** AIPMT 2004

A self-fertilizing trihybrid plant forms

(A) 8 different gametes and 64 different zygotes
(B) 4 different gametes and 16 different zygotes
(C) 8 different gametes and 16 different zygotes
(D) 8 different gametes and 32 different zygotes

Open solution

**Q166.** AIPMT 2003

Which one of the following traits of garden pea studied by Mendel was a recessive feature ?

(A) Green seed colour (B) Green pod colour
(C) Axial flower position (D) Round seed shape

Open solution

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**Q167.** AIPMT 2003

The genes controlling the seven pea characters studied by Mendel are now known to be located on how many different chromosomes ?

(A) Six (B) Seven (C) Five (D) Four

Open solution

**Q168.** AIPMT 2003

Two crosses between the same pair of genotypes or phenotypes in which the sources of the gametes are reversed in one cross, is known as : -

(A) Reciprocal cross (B) Reverse cross (C) Test cross (D) Dihybrid cross

Open solution

**Q169.** AIPMT 2003

Pattern baldness, moustaches and beard in human males are examples of : -

(A) Sex limited traits (B) Sex influenced traits
(C) Sex determining traits (D) Sex linked traits

Open solution

**Q170.** AIPMT 2003

Which one of the following conditions though harmful in itself, is also a potential saviour from a mosquito borne infectious disease ?

(A) Pernicius anaemia (B) Thalassaemia
(C) Leukemia (D) Sickle cell anaemia

Open solution

**Q171.** AIPMT 2003

Down's syndrome is caused by an extra copy of chromosome number 21. What percentage of offspring produced by an affected mother and a normal father would be affected by this disorder ?

(A) 50% (B) 100% (C) 75% (D) 25%

Open solution

**Q172.** AIPMT 2003

In Drosophila, the sex is determined by : -

(A) X and Y chromosomes
(B) Whether the egg is fertilized or develops parthenogenetically
(C) The ratio of number of X-chromosomes to the sets of autosomes
(D) The ratio of pairs of X-chromosomes to the pairs of autosomes

Open solution

**Q173.** AIPMT 2003

Christmas disease is another name for : -

(A) Hepatitis B (B) Sleeping sickness
(C) Haemophilia B (D) Down's syndrome

Open solution

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**Q174.** AIPMT 2003

Genetic map is one that

(A) shows the stages during the cell division

(B) shows the distribution of various species in a region

(C) establishes sites of the genes on a chromosome

(D) establishes the various stages in gene evolution

Open solution

**Q175.** AIPMT 2003

Genes for cytoplasmic male sterility in plants are generally located in

(A) nuclear genome

(B) cytosol

(C) chloroplast genome

(D) mitochondrial genome

Open solution

**Q176.** AIPMT 2003

When a cluster of genes show linkage behaviour they

(A) do not show independent assortment

(B) induce cell division

(C) do not show a chromosome map

(D) show recombination during meiosis

Open solution

**Q177.** AIPMT 2003

Which of the following discoveries resulted in a Nobel Prize?

(A) Recombination of linked genes

(B) Genetic engineering

(C) X-rays induce sex-linked recessive lethal mutations

(D) Cytoplasmic inheritance

Open solution

**Q178.** AIPMT 2003

The linkage map of X-chromosome of fruit-fly has 66 units, with yellow body gene ($y$) at one end and bobbed hair ($b$) gene at the other end. The recombination frequency between these two genes ($y$ and $b$) should be

(A) $\leq 50\%$

(B) $100\%$

(C) $66\%$

(D) $> 50\%$

Open solution

**Q179.** AIPMT 2002

Which of the following is the example of pleiotropy ?

(A) Thalassemea

(B) Haemophilia

(C) Sickle cell anaemia

(D) Colour blindness

Open solution

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**Q180.** AIPMT 2002

There are three genes a, b and c. The percentage of crossing over between a and b is 20%, b and c is 28% and a and c is 8%. What is the sequence of genes on chromosome

(A) a, b, c (B) b, a, c (C) a, c, b (D) None

Open solution

**Q181.** AIPMT 2002

On selfing a plant of $F_1$ generation with genotype "AABbCC",the phenotypic ratio in $F_2$ generation will be

(A) 1 : 1 (B) 3 : 1

(C) 9 : 3 : 3 : 1 (D) 27 : 9 : 9 : 9 : 3 : 3 : 3 : 1

Open solution

**Q182.** AIPMT 2002

A gene said to be dominant if : -

(A) It expressed only in heterozygous condition

(B) It express it's effect only in homozygous stage

(C) It never expressed in any condition

(D) It expressed both in homozygous and heterozygous condition

Open solution

**Q183.** AIPMT 2002

A diseased man marries a normal woman. They get three daughter and five sons. All the daughter were diseased and sons were normal. The gene of this disease is : -

(A) Sex linked dominant (B) Autosomal dominant

(C) Sex linked recessive (D) Sex limited character

Open solution

**Q184.** AIPMT 2002

Which of the following is a correct match -

(A) Sickel cell anaemia = X – Chromosome (B) Down Syndrome = $21^{st}$ Chromosome

(C) Haemophilia = Y – Chromosome (D) Parkinson Disease = X & Y Chromosome

Open solution

**Q185.** AIPMT 2002

Which of the following is the example of sex-linked disease?

(A) AIDS (B) Colour blindness (C) Syphilis (D) Gonorrhoea

Open solution

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**Q186.** AIPMT 2001

Two nonallelic genes produce the new phenotype when present together but fail to do so independently then it is called : -

(A) Polygene
(B) Complementary gene
(C) Non-complementary gene
(D) Epistasis

Open solution

**Q187.** AIPMT 2001

A and B genes are linked. What shall be genotype of progeny in a cross between AB/ab and ab/ab : -

(A) AABB and aabb
(B) None
(C) AaBb and aabb
(D) AAbb and aabb

Open solution

**Q188.** AIPMT 2001

When dominant and recessive alleles express themselves together it is called : -

(A) Dominance
(B) Pseudo dominance
(C) Co-dominance
(D) Amphidominance

Open solution

**Q189.** AIPMT 2001

Ratio of complementary genes is : -

(A) 9 : 3 : 4
(B) 9 : 7
(C) 12 : 3 : 1
(D) 9 : 3 : 3 : 4

Open solution

**Q190.** AIPMT 2001

Independent assortment of genes does not take place when : -

(A) Genes are located on non-homologous chromosome
(B) Genes are linked and located on same chromosome
(C) Genes are located on homologous chromosomes
(D) All the above

Open solution

**Q191.** AIPMT 2001

Sickle cell anaemia is induce to : -

(A) Change of Amino acid either a or b chain of Haemoglobin
(B) Change of Amino Acid in a chain of Haemoglobin
(C) Change of Amino acid in both a and b chain of Haemoglobin
(D) Change of Amino Acid in b chain of Haemoglobin

Open solution

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**Q192.** AIPMT 2001

Number of Barr bodies in XXXX female is

(A) 3 (B) 1 (C) 2 (D) 4

Open solution

**Q193.** AIPMT 2001

Male XX and female XY sometime occur due to

(A) Aneuploidy

(B) Deletion

(C) Transfer of segments in X and Y chromosomes

(D) Hormonal imbalance

Open solution

**Q194.** AIPMT 2001

Probability of four sons to a couple is : -

(A) $\frac{1}{16}$ (B) $\frac{1}{8}$ (C) $\frac{1}{4}$ (D) $\frac{1}{32}$

Open solution

**Q195.** AIPMT 2001

Extranuclear inheritance occurs in

(A) Killer *Paramecium* (B) Killer *Amoeba*

(C) *Euglena* (D) *Hydra*

Open solution

**Q196.** AIPMT 2001

In his experiment, Mendel obtained wrinkled pea. The wrinkling was due to deposition of sugar instead of starch. This happened due to the enzyme

(A) amylase (B) invertase

(C) diastase (D) absence of starch-branching enzyme

Open solution

**Q197.** AIPMT 2000

Due to the cross between TTRr × ttrr the resultant progenies showed how many percent plants tall, red flowered :

(A) 50% (B) 100% (C) 75% (D) 25%

Open solution

**Q198.** AIPMT 2000

According to mendelism which character is showing dominance ?

(A) Green colour in seed coat (B) Wrinkled seed

(C) Terminal position of flower (D) Green pod colour

Open solution

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**Q199.** AIPMT 2000

Mongolian idiots are due to trisomy in 21$^{st}$ chromosome is called :

(A) Klinefelter's syndrome
(B) Down's syndrome
(C) Turner's syndrome
(D) Triplex syndrome

Open solution

**Q200.** AIPMT 2000

In Drosophila the XXY condition leads to femaleness whereas in human beings the same condition leads to Klinefelter's syndrome in male. It proves :

(A) Y chromosome is active in sex determination in both human beings and Drosophila
(B) In Drosophila Y-chromosome decides femaleness
(C) Y chromosome of man have genes for syndrome
(D) In human beings Y chromosome is active in sex determination

Open solution

**Q201.** AIPMT 2000

Mutation generally produces

(A) recessive genes
(B) lethal genes
(C) polygenes
(D) dominant genes

Open solution

**Q202.** AIPMT 1999

A gene pair hides the effect of another gene. This phenomenon is called

(A) epistasis
(B) segregation
(C) mutation
(D) dominance

Open solution

**Q203.** AIPMT 1999

Hybridization between Tt $\times$ tt gives rise to the progeny of ratio :

(A) 1 : 1
(B) 1 : 2 : 1
(C) 1 : 2
(D) 4 : 1

Open solution

**Q204.** AIPMT 1999

A marriage between normal-visioned man and colourblind woman will produce offspring :

(A) 50% colourblind sons and 50% carrier daughters
(B) colourblind sons and carrier daughters
(C) normal males and carrier daughters
(D) colourblind sons and 50% carrier daughters

Open solution

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**Q205.** AIPMT 1999

Haemophilic man marries a normal woman. Their offspring will be :

(A) all boys haemophilic
(B) all normal
(C) all girls haemophilic
(D) all haemophilic

Open solution

**Q206.** AIPMT 1999

In human beings, multiple genes are involved in the inheritance of :

(A) skin colour
(B) phenylketonuria
(C) colour blindness
(D) sickle cell anaemia

Open solution

**Q207.** AIPMT 1999

Which of the following is the main category of mutation ?

(A) Somatic mutation (B) Genetic mutation (C) Zygotic mutation (D) All of the above

Open solution

**Q208.** AIPMT 1998

When a single gene influences more than one traits it is called

(A) pleiotropy
(B) epistasis
(C) pseudodominance
(D) none of these

Open solution

**Q209.** AIPMT 1998

If Mendel had studied the seven traits using a plant with 12 chromosomes instead of 14, in what way would his interpretation have been different?

(A) He would have mapped the chromosome
(B) He would have discovered blending or incomplete dominance
(C) He would not have discovered the law of independent assortment
(D) He would have discovered sex linkage

Open solution

**Q210.** AIPMT 1998

Crossing over in diploid organism is responsible for

(A) dominance of genes
(B) linkage between genes
(C) segregation of alleles
(D) recombination of linked alleles

Open solution

**Q211.** AIPMT 1998

How many different types of genetically different gametes will be produced by a heterozygous plant having genotype AABbCc?

(A) Two (B) Four (C) Six (D) Nine

Open solution

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**Q212.** AIPMT 1998

A woman with two genes (one on each 'X' chromosome) for haemophilia and one gene for colour blindness on the 'X' chromosomes marries a normal man. How will the progeny be?

Open solution

(A) All sons and daughters haemophilic and colourblind

(B) Haemophilic and colourblind daughters

(C) 50% haemophilic colourblind sons and 50% haemophilic sons

(D) 50% haemophilic daughters and 50% colourblind daughters

**Q213.** AIPMT 1998

Mental retardation in man, associated with sex chromosomal abnormality is usually due to

Open solution

(A) reduction in X complement

(B) increase in X complement

(C) moderate increase in Y complement

(D) large increase in Y complement

**Q214.** AIPMT 1998

Albinism is known to be due to an autosomal recessive mutation. The first child of a couple with normal skin pigmentation was an albino. What is the probability that their second child will also be an albino?

Open solution

(A) 100%

(B) 25%

(C) 50%

(D) 75%

**Q215.** AIPMT 1998

The formation of multivalents at meiosis in diploid organism is due to

Open solution

(A) monosomy

(B) inversion

(C) deletion

(D) reciprocal translocation

**Q216.** AIPMT 1998

Loss of an X-chromosome in a particular cell, during its development, results into

Open solution

(A) diploid individual

(B) triploid individual

(C) gynandromorphs

(D) both 'a' and 'b'

**Q217.** AIPMT 1997

A fruit fly is heterozygous for sex-linked genes, when mated with normal female fruit fly, the males specific chromosome will enter egg cell in the proportion

Open solution

(A) 3 : 1

(B) 7 : 1

(C) 1 : 1

(D) 2 : 1

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**Q218.** AIPMT 1997

A person with the sex chromosomes XXY suffers from

(A) gynandromorphism (B) Klinefelter's syndrome
(C) Down's syndrome (D) Turner's syndrome

Open solution

**Q219.** AIPMT 1997

Genetic identity of a human male is determined by

(A) sex-chromosome (B) cell organelles (C) autosome (D) nucleolus

Open solution

**Q220.** AIPMT 1996

When an animal has both the characters of male and female, it is called

(A) super female (B) super male (C) intersex (D) gynandromorph

Open solution

**Q221.** AIPMT 1996

The polygenic genes show

(A) different karyotypes (B) different genotypes
(C) different phenotypes (D) none of these

Open solution

**Q222.** AIPMT 1996

When two dominant independently assorting genes react with each other, they are called

(A) collaborative genes (B) complementary genes
(C) duplicate genes (D) supplementary genes

Open solution

**Q223.** AIPMT 1996

In which of the following diseases, the man has an extra X-chromosome?

(A) Turner's syndrome (B) Klinefelter's syndrome
(C) Down's syndrome (D) Haemophilia

Open solution

**Q224.** AIPMT 1996

A person whose father is colour blind marries a lady whose mother is daughter of a colour blind man. Their children will be

(A) all sons colour blind (B) some sons normal and some colour blind
(C) all colour blind (D) all daughters normal

Open solution

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**Q225.** AIPMT 1996

A genetically diseased father (male) marries with a normal female and gives birth to 3 carrier girls and 5 normal sons. It may be which type of genetic disease?

(A) Sex-influenced disease
(B) Blood group inheritance disease
(C) Sex-linked disease
(D) Sex-recessive disease

Open solution

**Q226.** AIPMT 1995

When two genetic loci produce identical phenotypes in *cis* and *trans* position, they are considered to be

(A) multiple alleles
(B) the parts of same gene
(C) pseudoalleles
(D) different genes

Open solution

**Q227.** AIPMT 1995

The phenomenon, in which an allele of one gene suppresses the activity of an allele of another gene, is known as

(A) epistasis
(B) dominance
(C) suppression
(D) inactivation

Open solution

**Q228.** AIPMT 1995

The most striking example of point mutation is found in a disease called

(A) Down's syndrome
(B) sickle cell anaemia
(C) thalassaemia
(D) night blindness

Open solution

**Q229.** AIPMT 1995

An abnormal human male phenotype involving an extra X-chromosome (XXY) is a case of

(A) Edward's syndrome
(B) Klinefelter's syndrome
(C) intersex
(D) Down's syndrome

Open solution

**Q230.** AIPMT 1994

The genes, which remain confined to differential region of Y-chromosome, are

(A) autosomal genes
(B) holandric genes
(C) completely sex-linked genes
(D) mutant genes

Open solution

**Q231.** AIPMT 1994

A child's blood group is 'O'. The parent's blood groups cannot be

(A) A and B
(B) A and A
(C) AB and O
(D) B and O

Open solution

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**Q232.** AIPMT 1994

Albinism is a congenital disorder resulting from the lack of which enzyme?

(A) Tyrosinase (B) Xanthine oxidase (C) Catalase (D) Fructokinase

Open solution

**Q233.** AIPMT 1994

The colour blindness is more likely to occur in males than in females because

(A) the Y-chromosome of males have the genes for distinguishing colours

(B) genes for characters are located on the sex-chromosomes

(C) the trait is dominant in males and recessive in females

(D) none of the above

Open solution

**Q234.** AIPMT 1993

Of both normal parents, the chance of a male child becoming colour blind are

(A) no

(B) possible only when all the four grand parents had normal vision

(C) possible only when father's mother was colour blind

(D) possible only when mother's father was colour blind

Open solution

**Q235.** AIPMT 1993

Mr. Kapoor has Bb autosomal gene pair and d allele sex-linked. What shall be proportion of Bd in sperms?

(A) Zero (B) 1/2 (C) 1/4 (D) 1/8

Open solution

**Q236.** AIPMT 1993

Which of the following is suitable for experiment on linkage?

(A) aaBB × aaBB (B) AABB × aabb (C) AaBb × AaBb (D) AAbb × AaBB

Open solution

**Q237.** AIPMT 1993

Of a normal couple, half the sons are haemophiliac while half the daughters are carriers. The gene is located on

(A) X-chromosome of father (B) Y-chromosome of father

(C) one X-chromosome of mother (D) both the X-chromosomes of mother

Open solution

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**Q238.** AIPMT 1993

Two dominant nonallelic genes are 50 map units apart. The linkage is

(A) cis type
(B) trans type
(C) complete
(D) absent/incomplete

Open solution

**Q239.** AIPMT 1993

A polygenic inheritance in human beings is

(A) skin colour
(B) phenylketonuria
(C) colour blindness
(D) sickle cell anaemia

Open solution

**Q240.** AIPMT 1993

Mendel studied inheritance of seven pairs of traits in Pea which can have 21 possible combinations. If you are told that in one of these combinations, independent assortment is not observed in later studies, your reaction will be

(A) independent assortment principle may be wrong
(B) Mendel might not have studied all the combinations
(C) it is impossible
(D) later studies may be wrong

Open solution

**Q241.** AIPMT 1993

Sex is determined in human beings

(A) by ovum
(B) at time of fertilization
(C) 40 days after fertilization
(D) seventh to eight week when genitals differentiate in fetus

Open solution

**Q242.** AIPMT 1992

The genotype of B blood group father of an O blood group child is

(A) $I^OI^O$
(B) $I^BI^B$
(C) $I^AI^B$
(D) $I^OI^B$

Open solution

**Q243.** AIPMT 1992

An allele is dominant if it is expressed in

(A) both homozygous and heterozygous states
(B) second generation
(C) heterozygous combination
(D) homozygous combination

Open solution

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**Q244.** AIPMT 1992

In a cross between AABB × aabb, the ratio of $F_2$ genotypes between AABB, AaBB, Aabb and aabb would be

(A) 9 : 3 : 3 : 1 (B) 2 : 1 : 1 : 2 (C) 1 : 2 : 2 : 1 (D) 7 : 5 : 3 : 1

Open solution

**Q245.** AIPMT 1992

Segregation of Mendelian factors (no linkage, no crossing over) occurs during

(A) anaphase I (B) anaphase II (C) diplotene (D) metaphase I

Open solution

**Q246.** AIPMT 1992

An organism with two identical alleles is

(A) dominant (B) hybrid (C) heterozygous (D) homozygous

Open solution

**Q247.** AIPMT 1992

A colour blind mother and normal father would have

(A) colour blind sons and normal/carrier daughters

(B) colour blind sons and daughters

(C) all colour blind

(D) all normal

Open solution

**Q248.** AIPMT 1992

Down's syndrome is due to

(A) crossing over (B) linkage

(C) sex-linked inheritance (D) nondisjunction of chromosomes

Open solution

**Q249.** AIPMT 1992

In human beings 45 chromosomes/ single X/XO abnormality causes

(A) Down's syndrome (B) Kinefelter's syndrome

(C) Turner's syndrome (D) Edward's syndrome

Open solution

**Q250.** AIPMT 1991

A man of A-blood group marries a women of AB blood group. Which type of progeny would indicate that man is heterozygous A?

(A) AB (B) A (C) O (D) B

Open solution

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**Q251.** AIPMT 1991

Multiple alleles control inheritance of

(A) phenylketonuria (B) colour blindness

(C) sickle cell anaemia (D) blood groups

Open solution

**Q252.** AIPMT 1991

The contrasting pairs of factors in Mendelian crosses are called

(A) multiple alleles (B) allelomorphs (C) alloloci (D) paramorphs

Open solution

**Q253.** AIPMT 1991

First geneticist/father of genetics was

(A) De Vries (B) Mendel (C) Darwin (D) Morgan

Open solution

**Q254.** AIPMT 1991

Mendel's last law is

(A) segregation (B) dominance

(C) independent assortment (D) polygenic inheritance

Open solution

**Q255.** AIPMT 1991

Blue eye colour is recessive to brown eye colour. A brown eyed man whose mother was blue eyed marries a blue-eyed women. The children will be

(A) both blue eyed and brown eyed 1 : 1 (B) all brown eyed

(C) all blue eyed (D) blue eyed and brown eyed 3 : 1

Open solution

**Q256.** AIPMT 1991

The allele which is unable to express its effect in the presence of another is called

(A) codominant (B) supplementary (C) complementary (D) recessive

Open solution

**Q257.** AIPMT 1991

RR (Red) *Antirrhinum* is crossed with white (WW) one. Offspring RW are pink. This is an example of

(A) dominant-recessive (B) incomplete dominance

(C) hybrid (D) supplementary genes

Open solution

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**Q258.** AIPMT 1991

A colour blind girl is rare because she will be born only when

(A) her mother and maternal grand father were colour blind

(B) her father and maternal grand father were colour blind

(C) her mother is colour blind and father has normal vision

(D) parents have normal vision but grand parents were colour blind

Open solution

**Q259.** AIPMT 1990

Cross between AaBB and aaBB will form

(A) 1AaBB : 1aaBB (B) all AaBB (C) 3AaBB : 1aaBB (D) 1AaBB : 3aaBB

Open solution

**Q260.** AIPMT 1990

In a genetic cross having recessive epistasis, $F_2$ phenotypic ratio would be

(A) 9 : 6 : 1 (B) 15 : 1 (C) 9 : 3 : 4 (D) 12 : 3 : 1

Open solution

**Q261.** AIPMT 1990

ABO blood group system is due to

(A) multifactor inheritance (B) incomplete dominance

(C) multiple allelism (D) epistasis

Open solution

**Q262.** AIPMT 1990

tt mates with Tt. What will be characteristic of offspring?

(A) 75% recessive (B) 50% recessive (C) 25% recessive (D) All dominant

Open solution

**Q263.** AIPMT 1990

In Down's syndrome of a male child, the sex complement is

(A) XO (B) XY (C) XX (D) XXY

Open solution

**Q264.** AIPMT 1990

Haemophilia is more common in males because it is a

(A) recessive character carried by Y-chromosome

(B) dominant character carried by Y-chromosome

(C) dominant trait carried by X-chromosome

(D) recessive trait carried by X-chromosome

Open solution

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**Q265.** AIPMT 1990

Which one is a hereditary disease?

(A) Cataract (B) Leprosy (C) Blindness (D) Phenylketonuria

Open solution

**Q266.** AIPMT 1990

Both husband and wife have normal vision though their fathers were colour blind. The probability of their daughter becoming colour blind is

(A) 0% (B) 25% (C) 50% (D) 75%

Open solution

**Q267.** AIPMT 1990

Bateson used the terms coupling and repulsion for linkage and crossing over. Name the correct parental of coupling type alongwith its cross over or repulsion.

(A) Coupling AABB, aabb; Repulsion AABB, aabb

(B) Coupling AAbb, aaBB; Repulsion AaBb, aabb

(C) Coupling aaBB, aabb; Repulsion AABB, aabb

(D) coupling AABB, aabb : Repulsion AAbb, aaBB

Open solution

**Q268.** AIPMT 1989

A normal green male maize is crossed with albino female. The progeny is albino because

(A) trait for a albinism is dominant

(B) the albinos have biochemical to destroy plastids derived from green male

(C) plastids are inherited from female parent

(D) green plastids of male must have mutated

Open solution

**Q269.** AIPMT 1989

Two linked genes a and b show 20% recombination. the individuals of a dihybrid cross between ++/++ × ab/ab shall show gametes

(A) ++ : 80 :: ab : 20 (B) ++ : 50 :: ab : 50

(C) ++ : 40 :: ab : 40 :: +a : 10 :: +b : 10 (D) ++ : 30 :: ab : 30 :: +a : 20 :: +b : 20

Open solution

**Q270.** AIPMT 1988

Which contribute to the success of Mendel?

(A) Qualitative analysis of data

(B) Observation of distinct inherited traits

(C) His knowledge of biology

(D) Consideration of one character at one time

Open solution

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**Q271.** AIPMT 1988

A family of five daughter only is expecting sixth issue. The chance of its beings a son is

(A) zero (B) 25% (C) 50% (D) 100%

Open solution

**Q272.** AIPMT 1988

Haploids are able to express both recessive and dominant alleles/mutations because there are

(A) many alleles for each gene

(B) two alleles for each gene

(C) only one allele for each gene in the individual

(D) only one allele in a gene

Open solution

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# Answer Key

| | | | | |
|---|---|---|---|---|
| 1. **B** | 2. **A** | 3. **C** | 4. **A** | 5. **A** |
| 6. **B** | 7. **A** | 8. **C** | 9. **D** | 10. **C** |
| 11. **A** | 12. **B** | 13. **B** | 14. **D** | 15. **D** |
| 16. **D** | 17. **D** | 18. **B** | 19. **B** | 20. **B** |
| 21. **C** | 22. **C** | 23. **A** | 24. **A** | 25. **B** |
| 26. **B** | 27. **D** | 28. **D** | 29. **A** | 30. **B** |
| 31. **B** | 32. **B** | 33. **D** | 34. **A** | 35. **A** |
| 36. **C** | 37. **C** | 38. **D** | 39. **A** | 40. **D** |
| 41. **C** | 42. **C** | 43. **A** | 44. **B** | 45. **C** |
| 46. **D** | 47. **D** | 48. **D** | 49. **D** | 50. **B** |
| 51. **A** | 52. **C** | 53. **C** | 54. **B** | 55. **C** |
| 56. **A** | 57. **D** | 58. **C** | 59. **D** | 60. **B** |
| 61. **A** | 62. **B** | 63. **A** | 64. **D** | 65. **A** |
| 66. **A** | 67. **B** | 68. **D** | 69. **B** | 70. **A** |
| 71. **B** | 72. **D** | 73. **C** | 74. **B** | 75. **C** |
| 76. **C** | 77. **A** | 78. **C** | 79. **A** | 80. **B** |
| 81. **B** | 82. **C** | 83. **D** | 84. **C** | 85. **C** |
| 86. **C** | 87. **B** | 88. **B** | 89. **D** | 90. **B** |
| 91. **B** | 92. **C** | 93. **C** | 94. **B** | 95. **A** |
| 96. **B** | 97. **B** | 98. **A** | 99. **C** | 100. **A** |
| 101. **B** | 102. **B** | 103. **A** | 104. **B** | 105. **A** |
| 106. **A** | 107. **B** | 108. **D** | 109. **D** | 110. **A** |
| 111. **D** | 112. **A** | 113. **C** | 114. **B** | 115. **A** |
| 116. **B** | 117. **B** | 118. **D** | 119. **C** | 120. **D** |
| 121. **A** | 122. **D** | 123. **B** | 124. **C** | 125. **B** |
| 126. **A** | 127. **C** | 128. **B** | 129. **D** | 130. **A** |
| 131. **A** | 132. **A** | 133. **A** | 134. **C** | 135. **B** |
| 136. **D** | 137. **A** | 138. **C** | 139. **A** | 140. **B** |
| 141. **D** | 142. **B** | 143. **D** | 144. **A** | 145. **A** |
| 146. **C** | 147. **A** | 148. **C** | 149. **B** | 150. **B** |
| 151. **A** | 152. **B** | 153. **D** | 154. **C** | 155. **A** |

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| | | | | |
|---|---|---|---|---|
| 156. **A** | 157. **A** | 158. **D** | 159. **C** | 160. **C** |
| 161. **A** | 162. **C** | 163. **D** | 164. **C** | 165. **A** |
| 166. **A** | 167. **D** | 168. **A** | 169. **B** | 170. **D** |
| 171. **A** | 172. **C** | 173. **C** | 174. **C** | 175. **D** |
| 176. **A** | 177. **C** | 178. **A** | 179. **C** | 180. **B** |
| 181. **B** | 182. **D** | 183. **A** | 184. **B** | 185. **B** |
| 186. **B** | 187. **C** | 188. **C** | 189. **B** | 190. **B** |
| 191. **D** | 192. **A** | 193. **C** | 194. **A** | 195. **A** |
| 196. **D** | 197. **A** | 198. **D** | 199. **B** | 200. **D** |
| 201. **A** | 202. **A** | 203. **A** | 204. **B** | 205. **B** |
| 206. **A** | 207. **B** | 208. **A** | 209. **C** | 210. **D** |
| 211. **B** | 212. **C** | 213. **B** | 214. **B** | 215. **D** |
| 216. **C** | 217. **C** | 218. **B** | 219. **A** | 220. **D** |
| 221. **C** | 222. **B** | 223. **B** | 224. **D** | 225. **C** |
| 226. **D** | 227. **A** | 228. **B** | 229. **B** | 230. **B** |
| 231. **C** | 232. **A** | 233. **B** | 234. **D** | 235. **C** |
| 236. **B** | 237. **C** | 238. **D** | 239. **A** | 240. **B** |
| 241. **B** | 242. **D** | 243. **A** | 244. **C** | 245. **A** |
| 246. **D** | 247. **A** | 248. **D** | 249. **C** | 250. **D** |
| 251. **D** | 252. **B** | 253. **B** | 254. **C** | 255. **A** |
| 256. **D** | 257. **B** | 258. **B** | 259. **A** | 260. **C** |
| 261. **C** | 262. **B** | 263. **B** | 264. **D** | 265. **D** |
| 266. **A** | 267. **D** | 268. **C** | 269. **C** | 270. **D** |
| 271. **C** | 272. **C** | | | |

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